Computational pipeline and evidence guide

Companion to Persistence of a certified scalar instability under dynamical interface matter — revision 6

27 October 2026

Abstract

An operational guide to the exact inputs, test dependencies, acceptance gates and saved outputs supporting the article. Repository-relative locators connect its conditional claims to interval certificates, nonlinear source/inverse records and the final observable budget. Portable saved-record checks are distinguished from fresh producer regeneration. A searchable companion catalog retains failed and superseded experiments.

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1 Purpose and reading route

This supplement describes how the computer-assisted premises of the article are assembled: the exact problem supplied to each stage, the calculation or check performed, the acceptance gates, the saved outputs, and the inputs passed to the next stage. It accompanies the article's mathematical appendices; it does not replace their proofs. The intended reader should be able to locate a claimed premise, inspect its input and acceptance contract, and determine which parts can be checked from the deposited records and which require a new producer execution.

The account is deliberately explicit about execution. An archive identity check establishes that the intended files were received. A saved-record check establishes specified identities, coverage or terminal inequalities. An algebra check establishes the displayed identities it evaluates. A producer regeneration computes fresh interval trajectories or source enclosures. These are different operations. A successful operation at one level is not reported as success at another.

The article's original revision 5 remains a preserved fallback. This guide concerns the **revision 6** claim, including its neutral nonlinear addition. The original revision 6 article is unchanged by preparation of this supplement. Its PDF has SHA-256

`388f2cd7ccbcf999f30ab361c0715cb3a65f3736c0404e4418f5fd775e205158`.

All locations below are **relative to the public pipeline repository root**, which is the private parent project's `pipeline/` submodule. No private parent checkout is required for the documented public record checks. The source/data snapshot is public-repository commit

`f135faa390af8cf7f497e96407a03cae052662e7`.

This supplement, its catalog and the revision 6 PDF are subsequent local additions at preparation time. The base commit identifies the scientific archive they describe; it does not contain these additions. A release should record the later publication commit as well. An owner-identifying repository URL is omitted from this anonymous reading copy. Repository availability and anonymous referee access must be resolved separately; a public owner URL does not provide anonymous access.

Read Sections 2–3 for the data flow and exact initial problem, Sections 4–7 for the stage contracts, and Section 8 for executable record checks. Section 9 identifies the remaining regeneration and review work. The searchable catalog provides the detailed historical inventory without turning every failed predecessor into assigned reading.

2 Repository map and claim-to-test flow

Public relative location	Role
`src/verify_saved_records.py`	Portable neutral saved-record checker; no producer imports.
`src/verify_l001_archive.py`	Streams all archived L001 files through SHA-256; checks file inventory.
`src/build_pipeline_catalog.py`	Generates the documentary catalog by reading JSON and parsing source text.
`src/historical/{background,phase,mass,kinetic}/`	Selected historical producer sources; preserved execution assumptions.
`src/l001/experiments/`	Relative symlink to `data/raw/l001/experiments/`; an entry point, not a second copy.
`tests/check_local_algebra.py`	Bounded symbolic and homogeneity checks using SymPy.
`data/raw/neutral-certificates/`	Background, phase, selected-mass and kinetic source/run packets.

Public relative location	Role
`data/processed/neutral-certificates/`	Anonymous inspection derivatives, source identities and manifest; some files differ from originals by recorded redaction.
`data/raw/l001/experiments/`	Full imported L001 experiment archive, including failed and superseded cases.
`data/processed/l001/ARCHIVE_MANIFEST.json`	Inventory and SHA-256 identities of 19,416 experiment files, totaling 14,040,303,990 bytes.
`data/processed/l001/CURRENT_RESULT.json`	Pointer to the accepted ninth-order complete observable budget and its audit.
`docs/pipeline-supplement/`	Generated test catalog, interpreted stage map and documentary validation records.
`manuscript/paper.pdf`	Preserved default revision 5 PDF.
`manuscript/revisions/20261027_revision_6/output/pdf/`	Generated revision 6 article and this companion PDF.

The repository contains Git LFS objects. A checkout containing pointer files is not a numerical-data checkout. Retrieve the objects before checking identities. The archive includes copied inputs and review snapshots; repeated bytes are not additional independent experiments. Git does not retain empty directories, and the inventory verifier treats absent historically empty directories separately from missing evidence files.

For compactness, define two location prefixes used in the stage cards:

- **NC** = `data/raw/neutral-certificates/`.
- **EX** = `data/raw/l001/experiments/`.

Thus `EX/20261027_ACTUAL_NINTH_FULL_BUDGET/RESULT.json` means the full repository-relative path obtained by substituting the EX prefix. These abbreviations are documentation notation, not shell variables.

Article conclusion	Computational input	Analytical bridge that must also be read
Complete neutral negative count, Proposition 1	B1 background/root and B2 zero-mass phase	Appendices J–N: physical reconstruction, domain, artificial-root removal, restricted index and wall continuation.
Exactly one negative mode after adding oscillators, Theorem 2	Same linked neutral background; B3 is a selected comparison mass	Appendix B: undivided oscillator rows, negative-axis elimination, monotone fixed point and positive norm.
Actual charged branch and growing linear direction	Neutral reference and nonresonant comparison; exact charged source definitions	Appendices C–D: compatible recursion, inverse/source bounds and convergence. No numerical finite-charge radius is supplied.
Canonical evolution and momentum control	B4 neutral kinetic anchor	Appendices E–G: constraints, common domain, corrected energy and compact-time small-charge continuation.

Article conclusion	Computational input	Analytical bridge that must also be read
Neutral nonlinear family, Theorem 3	N1–N5: seed, inverse, source bounds, wall identity and selected majorant	Appendix Q.1–Q.4: compatible harmonic induction, analytic convergence and recovery of the remaining wall equation.
Quantitative proper-clock departure and physical witness	N6–N9: finite coefficients, tails and full budget	Appendix Q.5–Q.6: uniform remainder inequalities and observable interpretation.

The charged and neutral nonlinear routes share an action and neutral anchor. Their expansion variables, rates and conclusions differ. The nonlinear packet is not a charged evolution experiment. The geometric comparison sector in Appendix P is an analytical substitution and wall/expansion calculation, not a numerical collapse or event-horizon producer.

3 Exact initial problem and conventions

3.1 Model parameters and launch

The six action parameters are exact rationals, not rounded decimal targets. They can be read without private files from

`NC/kinetic/inputs/14_21_14_04_MODEL.json`

under `model_exact`, and from the phase input binding under

`NC/phase/phase_regeneration/source/inputs/central_bindings.json`.

Parameter	Exact value
γ	20
λ_a	6755388606266369/1125899906842624
λ_b	−851862101055633/140737488355328
m^2	7385903388887613/18014398509481984
v_0	3602879701896397/36028797018963968
v_1	6520040733916207/72057594037927936

The lower wall is at proper radial coordinate zero, the warp is normalized there, and the exact shooting root belongs to

$$I_A = \left[\frac{456607025194779413}{4611686018427387904}, \frac{114151756532473697}{1152921504606846976}\right], \qquad I_D = \left[\frac{32727}{16384}, \frac{32729}{16384}\right].$$

For every enclosed launch A, use

$$W(0) = 0, \quad \Phi(0) = A, \quad T(0) = \gamma(A - v_0)/2, \quad H(0) = -[\lambda_a + \gamma(A - v_0)^2/2]/6,$$
$$k = H(0)^2 - 1 - T(0)^2/12 + m^2A^2/12.$$

The background equations are $W' = H$, $H' = -T^2/3 - ke^{-2W}$, $\Phi' = T$, and $T' = m^2\Phi - 4HT$. Here $a = e^W$, $E = e^{-W}$, $h = \sqrt{k}$; the prime denotes the proper radial derivative. The paper fixes $\ell = \kappa_5 = 1$. The interval parameter $k(A)$ is fixed along each trajectory. Replacing I_A by its midpoint, or recomputing k from an approximate evolving constraint, changes the declared problem.

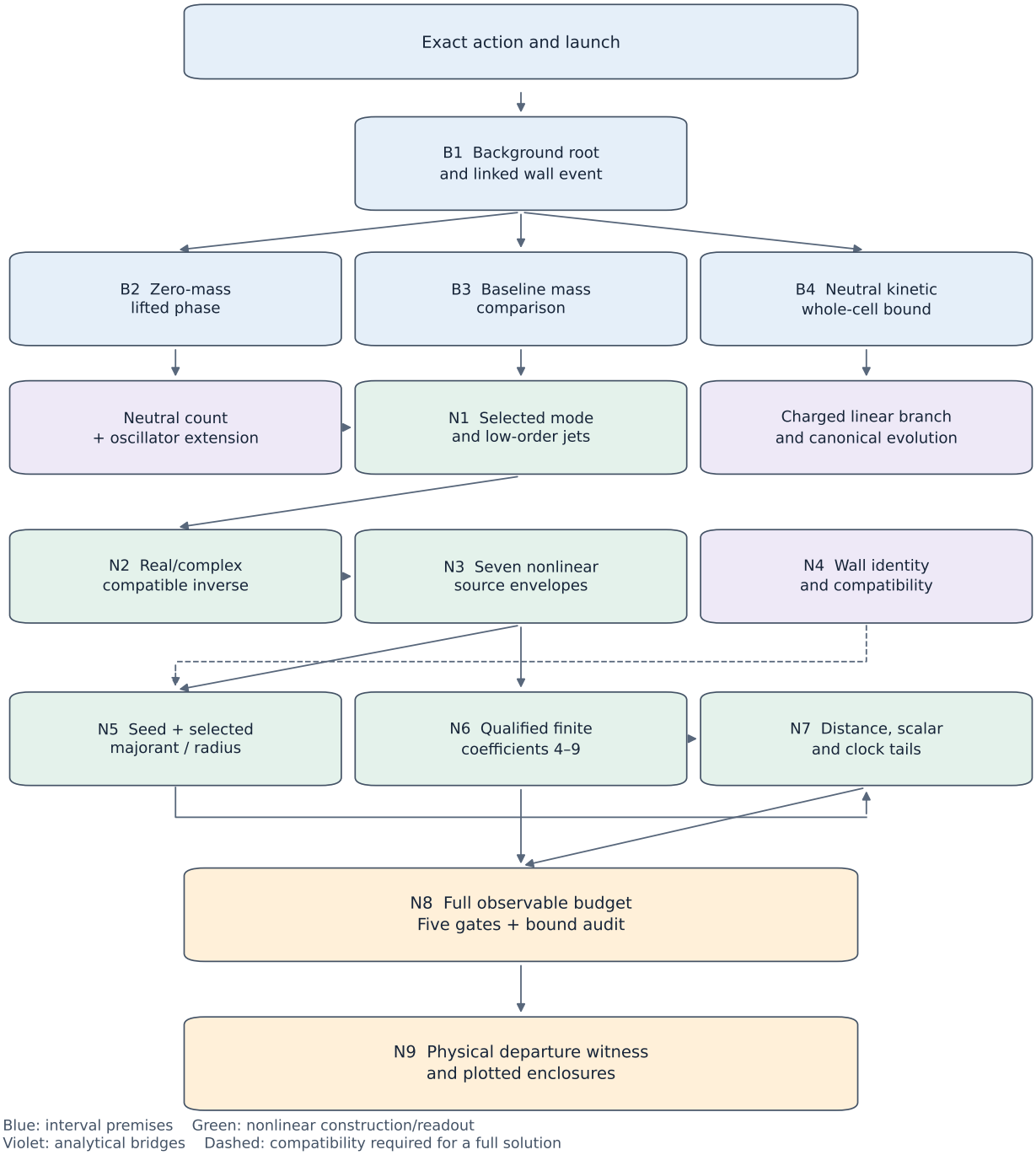


Figure 1: Functional flow of the article's evidence. Solid arrows denote mathematical consumption or an accepted readout; the searchable catalog resolves the more extensive bound provenance references.

Some original background sources spell parameters as decimal strings and convert them with `float(text).as_integer_ratio()`. Their intended values are the resulting binary rationals in the table, **not** exact decimal fractions. The catalog points to the original conversion code; a future portable adapter must compare the resulting rationals exactly before doing science.

The upper wall is a certified first transverse metric-junction event, followed by the scalar-junction shooting equation there. It is not a user-chosen final grid node. Every spectral/enclosure stage must cover the complete inherited uncertain event interval and the same identified exact root.

3.2 Distinct perturbation launches

B2, zero perturbative Hessians: at physical $\mu = 0$, use $\psi_a = 1$, $f_a = -12k/T'_a$, $g_a = -2T_a$. The background potential coefficient remains $\gamma = 20$. This is $L_a = g_a + 2T_a\psi_a = 0$; it does not set the background junction coefficient to zero. The lifted launch phase is $3\pi/2$.

B3, baseline finite-Hessian mass comparison: the prospective mass bracket is $[-0.00132, -0.00131]$, with the decimal endpoints interpreted exactly. Use $\psi_a = 1$, $B_0 = \gamma T_a/2 - T'_a$, $f_a = 3(\mu + 4k)/B_0$, and $g_a = \gamma f_a/2 - 2T_a$. This baseline has no surface oscillator. It must not be substituted for the oscillator-completed eigenmode in N1.

N1, selected oscillator-completed mode: wall Hessians are 20, $\omega_i^2 = \beta_i = \eta_i = 1$, so the oscillator denominator is $2 - \mu/a_i^2$, not $1 - \mu/a_i^2$. Both currents vanish. Normalize $p_1(a) = 1$, set both first graph coefficients to zero and take

$$\mu_* = -r(r + 3h), \qquad \eta = \epsilon e^{rt}, \qquad r > h.$$

The lower effective Hessian is $\Gamma_a = 21 - 1/(2 - \mu_*)$, since $a_a = 1$. The selected-mode bracket, lower launch and parameter enclosures are preserved in

`EX/20261025T225750Z_AB_MECHANISM_ALLOCATION/T2R1_AB_SELECTED_MODE/`

and in

`EX/20261026_GROWING_INVERSE_READINESS_R1/INITIAL_primary.json`

with its independent counterpart. The second-order initializer reconstructs these from the exact background and checks their overlap and constraints. The analytic seed stage additionally checks exact native initial-state/parameter identity against these saved records. Higher coefficients are the compatible forced solutions; arbitrary homogeneous additions are not admitted. There is no generic random initial condition or random seed for this declared family.

The charged construction instead uses $z = \delta^2 e^{-6ht}$, independent fixed charge labels and fixed phase constants when asserting full-variable uniqueness. The historical model record's unit oscillator parameters and the selected denominator 2 are consistent: the denominator includes $\omega_i^2 + \beta_i$.

4 Neutral foundational certificates: B1–B4

4.1 B1 — exact background root, event and geometry

Question and inputs. Does the exact two-wall action admit the linked local root with positive warp/curvature and the required endpoint geometry? Inputs are the six rationals, frozen narrow/wide shooting boxes and the original event residuals in Appendix O.1. The tight root image is an output, not an adjustable shooting target.

Folder and source. Use `NC/background/background_regeneration/`. Source bindings are in `SOURCE_BINDINGS.json`. The primary wrapper is `source/primary/confirm_32768.py`, importing `validated_sensitivity_newton.py`. The independently written route is `source/independent/independent_d1_replay_r1.py`, importing `independent_d1_replay.py`. Selected copies also occur under `src/historical/background/`.

Operation. The primary wrapper selects 768-bit Arb and 32,768 subdivisions of $[0, 2]$. It encloses background and launch sensitivities and applies interval Newton to the event-composed residual. Its original sensitivity integrator and the independent Taylor/Picard integrator are different implementations. The independent R1 uses 8,192 and 16,384 subdivisions, 768-bit arithmetic and its own fifth-order remainder construction. Do not replace the different algorithms by a claim of identical numerical order.

Acceptance gates. Required predicates include strict interval-Newton inclusion in the frozen launch box, derivative excluding zero, nonempty first-event sign regions, transverse event, self-containing cell bounds, positive warp/curvature and the stated geometry. Common-box uniqueness identifies the two root enclosures. An overlap of approximate roots alone does not identify the same exact background.

Outputs and next consumers. `output/primary_RESULT.json` supplies `finest_narrow_result`, its exact Newton image and event bracket; `output/independent_RESULT.json` supplies the independent local root/geometry. Runtime witnesses and `output/EXECUTION.json` identify the 18 October executions. `NC/background/FINAL_INDEPENDENT_REVIEW.md` records the historical internal disposition. The linked I_A, I_D feed B2, B3, B4 and the selected-mode construction.

Replay boundary. The current standard-library verifier checks saved strict inclusion and launch/event identities. It does not integrate the background, recompute event sensitivity or reconstruct the omitted cell proofs. Historical wrappers retain runtime and filesystem assumptions.

4.2 B2 — zero-mass phase and remaining-band exclusion

Inputs and folder. B1's full I_A, I_D, exact model, physical $\mu = 0$, and zero perturbative Hessians. Use `NC/phase/phase_regeneration/source/`, especially `test1_phase_v3.py`, `inputs/central_bindings.json` and the two files in `engine/`.

Operation. Two regular scalar IVP engines use 768-bit Arb and one arithmetic thread. Their steps are 2/8192 and 2/16384. Each cell encloses the entire phase chart for (f, L), then propagates its lift from the previous known phase. Cells intersecting I_D are clipped to the full event intersection, and every intersection is included in the final hull.

Acceptance gates. The launch constraint and signs must hold. Each full cell must admit a half-plane chart avoiding the origin. The outward lifted angle must be less than π from the reference midpoint; endpoint and event charts must be compatible with that chart. There must be no event-cover gap, and both implementations must exclude threshold contact and agree on the integer count. The strict final band is

$$1 < \theta_b(0)/\pi < 3/2, \qquad \theta_b(0)/\pi \in \left[\frac{577838035}{536870912}, \frac{577858823}{536870912}\right] \quad \text{(displayed primary enclosure)}.$$

Outputs and use. `source/test1_v3/primary_result.json`, `independent_result.json` and their traces are the saved readouts. `EXECUTION_SOURCE_BINDINGS.json`, `output/` runtime records and `NC/phase/FINAL_INDEPENDENT_REVIEW.md` identify execution and review. The phase bound combines with the analytical restricted count and soft-corner injection to give the complete neutral count. It does not establish the background root independently.

Replay boundary. Checking saved event coverage and phase inequalities cannot reconstruct the original Picard boxes, Taylor remainders, elementary-function bounds or rigorous lift. Shared Arb and shared chart mathematics preclude a claim of arithmetic independence.

4.3 B3 — selected baseline mass and nonresonant comparison

Inputs and folder. Use `NC/mass/CONTRACT.md`, `FREEZE_RECORD.json`, `BINDINGS.json`, and `output/`. Inputs are B1's exact uncertainty, the baseline launch above and the frozen negative mass bracket. The active sources are `primary_interval.py`, `independent_interval.py` and `runner.py`.

Operation and gates. Four endpoint integrations run in the predeclared order: primary low/high,

then independent low/high. Arb precision is 768 bits; steps are 2/8192 and 2/16384. Each engine uses strict Picard containment, a degree-four Taylor polynomial and whole-cell order-five remainder. Finite enclosures, positive warp and constraint/first-integral consistency are required. Every event-intersection determinant range is included. Both engines must give strict opposite endpoint signs and compatible same-endpoint enclosures, with the entire bracket below $-4k$.

The contract distinguishes a further whole-mass-strip positivity/separation qualification from endpoint existence. Its failure cannot be converted into a separation claim from endpoint profiles. Original checkpoints canonicalize native Arb radii outward before storing and continuing; recovery must preserve the canonicalized state and verify guards. These controls do not prove arbitrary raw Arb serialization is lossless.

Outputs and use. `output/{primary,independent}_{low,high}.json` contain the sign/event records. `INDEPENDENT_PROOF_AUDIT.md`, `INDEPENDENT_AUDIT.json` and `POSTRUN_REVIEW.md` supply historical review scope. Appendix O.6 explains how the baseline mass bounds provide a nonresonant comparison. The selected oscillator-completed seed is separately characterized in N1; it is not this baseline mass certificate relabeled.

4.4 B4 — all-mode neutral canonical kinetic anchor

Inputs and folder. Use `NC/kinetic/INTERVAL_CERTIFICATE_PLAN.md`, `INPUTS.json`, `WEIGHTED_ALL_MODE_PROOF.md`, `certificate/` and `terminal_review/`. Inputs are the inherited exact root boxes and the analytical completion of the canonical form. This stage does not re-solve the root.

Operation. A six-state proper-radial enclosure propagates (W, H, Φ, T, E, q), with $E' = -HE$, $q' = E$. It uses Python integers with 192 fractional binary bits and directed rounding after every multiplication/division; it does not use Arb or sampled numerical matrices. The step is 1/8192, with a final shortened step, degree-four Taylor polynomial, whole-cell fifth-order remainder and at most 32 Picard iterations. There are **16,365 unique physical cells**. A separate 256-cell recovery replay is a control, not additional radial coverage.

Acceptance gates. Positive E, $0 < k < 1/3000$, $E < 8$, $m^2\Phi^2 < 1$, positive event width and the diagnostic sign of H are checked throughout their declared domains. The first integral must contain zero. Upper residuals over the enclosing rectangle must contain zero as consistency checks; they do not certify that every rectangle point solves both wall equations. The exact bound is

$$\Delta_{\mathrm{kin}} \le 12091872/47265625 < 1/2.$$

Outputs and use. `certificate/RESULTS.json`, `LATEST.json`, `checkpoints/` and compressed `cells/` bind state and whole-collar/event accumulation. `TERMINAL_ACCEPTANCE.json` records acceptance. With Appendix G's exact reduction and neutral-time density rescaling, the certificate gives $\mathscr{K}_0 \ge \mathscr{S}_0/2$. Small-charge continuation and canonical evolution remain analytical common-domain arguments.

Recovery and replay. Durable chunks contain 256 cells, have deterministic gzip identities, and are committed through an atomic latest pointer. Committed corrupt data or changed source must be rejected before any new cell; uncommitted temporary files are not trusted. The public checker rebuilds stored hulls and coverage, not Taylor derivatives or interval trajectories. No finite-charge numerical kinetic range is supplied.

5 Neutral nonlinear construction: N1–N5

5.1 N1 — selected mode, radial jets and proper-clock readout

Question. Which exact normalized growing direction is continued, and which low-order observable coefficients belong to it? The selected oscillator-completed mode is in `EX/20261025T225750Z_AB_MECHANISM_ALLOCATION/T2R1_AB_SELECTED_MODE/`. The later folders `20261026_GROWING_MODE_QUADRATIC_R1`,

20261026_GROWING_MODE_CUBIC_BULK, 20261026_GROWING_MODE_CUBIC_WALL, 20261026_GROWING_MODE_CUBIC_FLOW, 20261026_GROWING_MODE_CUBIC_AFFINE and 20261026_GROWING_MODE_CLOCK_READOUT are all under EX.

Inputs and computation. The forced quadratic route retains exact parameter intervals, $p_1(a) = 1$, independent wall oscillators and moving interfaces. Primary/independent radial steps are 1/4096 and 1/8192, with 768-bit Arb. Its source/constraint controls and probe are separate from production. Cubic bulk, wall and affine pullback qualifications feed the cubic flow and readout. The flat reference interval and wall traces must be the same exact ones; spatial and total wall derivatives cannot be interchanged.

Gates. The selected-mode classification, lower divisors, initial constraints, first integral, wall residuals and complete uncertain-event coverage must pass. Production requires its admitted controls and probe. Source, input and runtime identities are checked against frozen parents. The declared rate bound $h < r < 11h/10$ is checked. A failed predecessor is preserved as a failed case, and R1 is a separate packet.

Outputs. Initial records, cell/chunk records, endpoint reconstructions and the coefficient readouts are saved in the corresponding folders. 20261026_GROWING_MODE_CLOCK_READOUT/RESULT.json contains both engine arms and values, including d_1, d_2, d_3, r, lapse coefficients and separation-reparameterized normal-form coefficients. These feed analytic seed controls, finite higher-order enclosures and the final observable budget.

Convention that matters. The proper-clock amplitude coefficient is $F_2 = r(d_2 - d_1)$, since the first inverse-lapse coefficient is -1. A field named proper_quadratic_rate in the historical readout belongs to a different normal form after reparameterizing by separation. It must not be inserted as F_2 in the amplitude series.

5.2 N2 — real all-order inverse and complex analytic collar

Folders. EX/20261026_GROWING_NEUTRAL_ALL_ORDER_INVERSE/ and EX/20261026_NEUTRAL_ANALYTIC_INVERSE/; inspect DERIVATION.txt, active Python source, BINDING.json, RESULT.json, TERMINAL.json and the included audits. Readiness and metric/component packets are identified in their bindings and the catalog.

Inputs. The selected background and seed normalization, real whole-cell coefficient bounds, $r > h$, exact domain and the compatible forced harmonic parameter $\mu_n = -nr(nr + 3h)$, $n \geq 2$.

Operation and gates. The constraint denominator has a positive n^2 lower bound. After $f = E^2 y$, finite low harmonics receive separate potential/Robin tests, and a positive high-order barrier supplies a uniform inverse. The complex stage encloses one common background collar, prevents scaled-denominator zeros and selects one smaller stadium on which the positive continuation matrix is a strict contraction. Primitive factors $\sigma/(p+1)$, $\sigma^2/[p(p+1)]$, with $p = 2n$, pay derivative weights. The metric is continued through its first-order constraint and a Nagumo estimate. This is not inversion at the resonant first mode.

Outputs and use. Exact rational inverse constants, finite-order bounds, collar radii and gate records feed the seven-source majorant and its refinements. The first packet uses terminal keys pass and scientific_certificate; later packets often use pass_status. A reader should inspect the schema rather than assume a single universal producer flag. The analytical completeness of compatible-source inversion remains a proof premise; positive stored constants alone cannot establish it.

5.3 N3 — seven source maps and their response bounds

Folders. Begin with EX/20261026_NEUTRAL_SOURCE_MAJORANT/DERIVATION.txt, its source and result. The four bulk rows are trace, mixed, Klein–Gordon and radial constraint. The three wall rows are tracefree Israel/graph, scalar-normal junction and oscillator. The FIELD_SPECIFIC, MIXED_SOURCE, SCALAR_RADIAL_SOURCE, TRACEFREE_SOURCE, SCALAR_NORMAL_WALL and ISRAEL_OSCILLATOR_WALL budget packets are enumerated by exact names and dependencies in the catalog.

Inputs and operation. All fields are pulled back with $s = u+X$, $j = 1+X_u$, $\mathcal{D} = r\eta\partial_\eta - (r\eta\partial_\eta X)j^{-1}\partial_u$ and $\mathcal{L} = j^{-1}\partial_u$. Backgrounds are composed at $u + X$, so $\mathcal{D}$ annihilates them. Bulk jet weights and wall primitive-degree payments bound each compatible lower-order residual. Each source loses its own constant and linear jets only after inputs are truncated below the target order. The selected inverse-response map is then applied to the seven loads.

Gates. Positive composition denominators, common collar gaps, admissible boost/graph size, source coefficient domination and compatible derivative/harmonic payments are essential. Wall acceleration, oscillator kinetic stress, boost friction and full transport are retained. The coarse explanatory envelopes in Appendix Q demonstrate a source-control mechanism. They are not the numerical envelopes from which the final sharpened K was obtained.

Outputs and use. Source envelopes, response constants and positive component load budgets feed N5. The equation-to-envelope correspondence of the sharpened seven-source bounds and the analytic inverse remain the principal scientific review target. This supplement locates that target; it does not independently certify all original Einstein/interface terms.

5.4 N4 — compatibility and recovery of the unsolved wall row

Folders. `EX/20261026_GROWING_NEUTRAL_COMPATIBILITY/`, `EX/20261026_NEUTRAL_WALL_IDENTITY/`, and `EX/20261026_NEUTRAL_TWO_SIDED_DEPARTURE/`.

Inputs and operation. The exact action residuals, moving-wall normal, scalar junction and oscillator equation are combined with bulk Bianchi and wall Codazzi/material-exchange identities. The wall calculation compares a full five-dimensional Christoffel/Ricci route with a warped-product route. Positive nonzero harmonic rate disposes of the remaining scalar Israel residual after the solved rows vanish; the zero harmonic is fixed separately.

Gates and outputs. Symbolic identities and stated sign/nonzero assumptions must hold. `RESULT.json` and the terminal/audit records distinguish these exact checks from integration. Their role is sufficiency: convergence of a convenient subset would not by itself give a solution of every action equation. The two-sided departure argument applies to the compatible family and its physical wall witness, not arbitrary Cauchy data or an eventual collapse outcome.

5.5 N5 — component seed, selected ray and convergence radius

Inputs. N1's full first-mode cell archives, N2's real/complex response bounds, N3's seven-source load estimates, and N4's compatibility identity.

Seed stage. `EX/20261026_NEUTRAL_ANALYTIC_SEED/` performs a fresh full radial replay in its historical run and certifies analytic continuation of the resonant first mode. Its original steps are 1/4096, 1/8192, 768-bit Arb, with complete-domain and native-initial-identity gates. `EX/20261027_COMPONENT_SEED_CERTIFICATE/` then reads those saved cells without a new ODE, reconstructs component maxima and continuation with a positive matrix inverse, and checks exact parent replay. Its retained bulk seed is 1215, graph seed zero and summed oscillator seed at most 1/8.

Conditioning/refinement route. The following packet sequence identifies changes to proof bounds, not new physical parameters. Dates alone are not dependency evidence: use each `BINDING.json`, the source's authenticated parent calls and the catalog. Several stages reference entire ancestral closures.

Packet suffix under EX	Purpose and gate retained
`20261026_COMPONENT_ROUTED_CERTIFICATE`	Separate scalar value, gradient and curvature responses; exactly replay the parent.
`20261027_SEPARATE_REAL_ANCHOR_CERTIFICATE`	Preserve distinct real and complex response constants.

Packet suffix under EX	Purpose and gate retained
`20261027_HARMONIC_MATCHED_INVERSE` and `20261027_HARMONIC_MATCHED_CERTIFICATE`	Match complete source/response pairs across finite harmonic cases and high-order ranges.
`20261027_SPLIT_FORCING_BOUNDARY_CERTIFICATE`	Separate interior load from boundary load before recombining the physical solution.
`20261027_BACKGROUND_COMPLEX_FACTORIAL_CERTIFICATE`	Retain background composition and factorial information in the appropriate response/source lanes.
`20261027_TYPED_KG_CERTIFICATE_R1`	Use the actual derivative type and enclosed jets for the Klein–Gordon background composition.
`20261027_BULK_DEPENDENCY_CERTIFICATE_R1`	Remove absent explicit oscillator dependence from bulk sources while retaining coupled response.
`20261027_JUNCTION_DEPENDENCY_CERTIFICATE`	Keep exactly linear oscillator/scalar coupling in the response, rather than invent a higher-order source.
`20261027_COMPONENT_SEED_CERTIFICATE`	Reuse saved first-mode cells; certify the component seed without new integration.
`20261027_DENOMINATOR_FACTORIAL` and `20261027_MASS_RETAINING_F_CERTIFICATE`	Preserve denominator/factorial and harmonic mass information; check old certificate replay.
`20261027_FINAL_FIXED_GRID_CERTIFICATE`	Retain all 1123 parent rows; append a fixed bounded refinement on 20 rays, at most 1723 rows; no adaptive iteration.

Other intermediate refinements, diagnostics and superseded candidates are retained in the full catalog. The selected `refined` lane of the final fixed-grid result is operative; a baseline lane or an earlier smaller radius is not silently substituted.

Final gates and outputs. `EX/20261027_FINAL_FIXED_GRID_CERTIFICATE/RESULT.json` records the candidate grid, source budgets, response weights, baseline replay, refined comparisons and terminal gates. The selected values are

$$\ell_g = 1/256, \quad \ell_c = 1, \quad a_0 = 21/2305843009213693952, \quad K = 109802048057794951.$$

$$s_1 = 1215, \qquad R_0 = (96K)^{-1},$$

$$\rho = \frac{R_0}{8s_1} = \frac{1}{1024584870836896246771 20}, \qquad Y = \frac{513R_0}{256} = \frac{171}{899498377689456238592}.$$

The scalar majorant is $w = s_1\zeta + Kw^2/(1-Kw)^2$. Strict self-mapping and contraction, followed by the inner-collar factor, give the stated sufficient radius. The graph/oscillator ray weights condition this bound; they are not physical energy fractions. A better radius alone does not imply a better proper-clock observable budget. That requires N6–N9.

6 Finite coefficients, tails and final acceptance: N6–N9

6.1 N6 — qualified coefficients through ninth order

These stages bound coefficients on the same analytic family, rather than evolve a finite-amplitude trajectory. The exact `DERIVATION.txt`, pins, source and audit of each packet specify which coefficients are signed and which are only absolute bounds.

Order	Actual source-enclosure folder under EX	Qualifications to inspect
4	20261027_QUARTIC_REAL_SOURCE_ENCLOSURE	20261027_QUARTIC_BULK_QUALIFICATION_R2, 20261027_QUARTIC_WALL_QUALIFICATION; actual clock readout in 20261027_ACTUAL_QUARTIC_CLOCK_CERTIFICATE.
5	20261027_QUINTIC_REAL_SOURCE_ENCLOSURE	20261027_QUINTIC_BULK_QUALIFICATION, 20261027_QUINTIC_WALL_QUALIFICATION; production aggregate is named production_RESULT.json.
6	20261027_SIXTH_REAL_SOURCE_ENCLOSURE	20261027_SIXTH_BULK_QUALIFICATION, 20261027_SIXTH_WALL_QUALIFICATION; separate actual-clock certificate.
7	20261027_SEVENTH_REAL_SOURCE_ENCLOSURE	20261027_SEVENTH_BULK_QUALIFICATION, 20261027_SEVENTH_WALL_QUALIFICATION, 20261027_SEVENTH_JET_QUALIFICATION.
8	20261027_EIGHTH_REAL_SOURCE_ENCLOSURE	20261027_EIGHTH_BULK_QUALIFICATION_R1, 20261027_EIGHTH_WALL_QUALIFICATION, 20261027_EIGHTH_JET_QUALIFICATION.
9	20261027_NINTH_REAL_SOURCE_ENCLOSURE	20261027_NINTH_BULK_QUALIFICATION_R2, 20261027_NINTH_WALL_QUALIFICATION, 20261027_NINTH_JET_QUALIFICATION.

Inputs and gates. Each stage authenticates its qualified bulk/wall expressions, inherited lower coefficients, selected parameters, domain and finite-order inverse. Parameter identity with each arm's INITIAL record is checked. Moving-endpoint transport uses qualified spatial jets; differentiating an unsigned sup-norm bound is not an admissible replacement. Omission-sensitivity controls and polynomial convolution checks qualify the retained terms. For ninth order, the cubic spatial derivative of the first-order density contributes through the fourth graph shift and must remain. The two arm enclosures must be compatible; their overlap is not an independent accuracy estimate.

Outputs and flow. ARM_primary.json, ARM_independent.json, source/load records and aggregate results feed subsequent orders and the complete readout. Unsigned ninth-order bounds do not supply a ninth-order sign or field profile. The distinction between an earlier generic ninth-order enclosure and the later actual-source bound matters: the final complete budget authenticates and replaces the intended coefficient only.

6.2 N7 — distance, inverse-lapse and unknown tails

Folders and inputs. Use EX/20261027_DEGREE16_DISTANCE_MAJORANT/, 20261027_DEGREE24_DISTANCE_MAJORANT/, 20261027_FINITE_INVERSE_LAPSE24/ and 20261027_POSITIVE_INVERSE_TAIL_QUALIFICATION/. They consume the analytic radius/aggregate, lower finite coefficients and the exact positive-series operations.

Operation and gates. Finite distance and inverse-lapse series, their cross terms, and positive unknown tails are accounted for separately. Coefficients and denominators must meet positivity/domain requirements. Unknown high-order distance, Euler derivative, wall scalar and scalar derivative tails are

retained; a truncated polynomial is not presented as the solution. Inverse-lapse inversion needs its own control, since a distance tail alone does not control the proper clock.

New evaluation point. The final route uses `EX/20261027_NINTH_THREE_QUARTER_TAIL/` and `EX/20261027_NINTH_THREE_QUARTER_INVERSE_TAIL/`. They provide freshly qualified bounds at both $q = 1/2$ and $q = 3/4$, where $|\eta| = q\rho$. A bound valid at half-radius is not extrapolated upward. Positive high-order powers permit the declared downward scaling from the new star point to smaller amplitudes.

Outputs. `RESULT.json` under each packet contains the exact evaluation points, constants, selected tails and gates. The source and terminal pins tie these to their inputs. The final budget consumes distance/scalar/Euler tails and inverse-lapse tails together.

6.3 N8 — complete observable budget and audit

Folder. `EX/20261027_ACTUAL_NINTH_FULL_BUDGET/`; source `check.py`, contract `DERIVATION.txt`, pins `PINS.json`, records `BINDING.json`, `RESULT.json`, `TERMINAL.json`.

Inputs. The accepted eighth-order complete budget, actual ninth-source enclosure, freshly qualified three-quarter distance/scalar tails and inverse-lapse tails, plus their bound ancestors and audits. The producer exactly replays the old eighth-order bracket and old F_9 accounting before changing the intended ninth coefficient. The physical radius, normalization and gate definitions remain fixed.

Five acceptance gates. For every amplitude in the protected subdisk, the complete positive upper bounds must satisfy:

1. Proper-clock remainder from cubic and higher terms $\le 10|\eta|^2$.
2. Nonlinear distance remainder $\le 25|\eta|$.
3. Wall-scalar remainder $\le |\eta|/40000$.
4. Euler derivative of that scalar remainder $\le |\eta|/40000$.
5. Opening/domain inequalities: $|\eta| < 1/512$ and $25|\eta|/64 - 60|\eta|^2 > 0$ at the positive test amplitude.

Nonnegative coefficient/tail bounds make the documented downward extension valid. The source checks old half-radius non-worsening, unchanged radius, new-star identity, parent/audit binding and before/after runtime identities. Gate failure would retain a smaller admitted bracket or an unresolved classification under the frozen procedure; it would not justify changing tolerances after seeing results.

Recorded outcome. All five gates pass at the prospectively tested endpoint

$$q_* = 3/4, \qquad \eta_* = \frac{1}{136611316111586166236160} \simeq 7.32004 \times 10^{-24}.$$

Gate	Saved ratio to its allowed budget at η_*
Clock	0.00168301813146
Distance	$4.45248014775 \times 10^{-22}$
Scalar	$1.77383442087 \times 10^{-5}$
Scalar derivative	$7.31750302073 \times 10^{-5}$

The classification is `PREDECLARED_THREE_QUARTER_ENDPOINT_ADMITTED`, with zero bisection iterations because the tested upper endpoint already passed. This is not a maximal physical-amplitude estimate.

Identity and review. The result SHA-256 is

`f42c5d1cfd8ce5f3617e700faf6a6c3269a0eaa3d9909959ef737d35338f8b73`;

the terminal SHA-256 is

`2a0f983f7ac60e5359c48b5a3219136347935b1ac6d5bcd0c26fb96fbd0d9642`.

The internal audit is `EX/20261027_QUARTIC_REAL_SOURCE_AUDIT/ACTUAL_NINTH_FULL_BUDGET_AUDIT.json`. It records PASS and 7264 checks, bound to that terminal/result. It is an internal source/record audit using shared inputs and arithmetic; its check count is not a count of independent physical experiments or a proof of every underlying source-domination premise.

6.4 N9 — observable interpretation and Figure 3

The final readout controls proper separation and the wall scalar, including the latter's proper-time derivative. The nonzero wall-scalar witness cannot be manufactured by a coordinate redefinition. The amplitude expansion, proper-clock residual and separation-reparameterized normal form are kept distinct.

Figure 3 is rendered from saved coefficient/budget data. It shows an enclosure and a remainder-budget comparison, not time-stepped nonlinear trajectories. Its plotting source, 129 dyadic plotting knots, selected exact input identities and numerical rendering settings are preserved in the private revision's `figures/nonlinear_departure_source/`, with the publication renderer under `tools/render_paper_figure.py`. The current public scientific archive supports the numerical statements; **the article figure's renderer/CSV have not yet been deposited as public pipeline assets**. That is an explicit figure-regeneration gap. This supplement's own data-flow figure has separate source in its private build folder.

The current evidence supports local neutral departure on the declared compatible family. It supplies neither a finite-charge nonlinear theorem nor a prediction of the eventual geometry. Its small sufficient amplitude is a quantitative limitation of the certificate, not an established threshold for observable astrophysical change.

7 Catalog, provenance and preserved failures

`docs/pipeline-supplement/TEST_CATALOG.md` is the searchable human inventory. Its JSON companion has one entry for every one of the 210 top-level L001 archive folders plus 107 eligible immediate child test folders: **317 entries**. Immediate children qualify when they are named test folders or belong to the historical event-horizon collection and contain a contract or terminal/result record. Deeper copied-input/review trees remain in the archive manifest and source index; they are not counted as new test executions.

For each catalog entry the JSON supplies:

- folder and purpose excerpt with its source location;
- top-level producer/helper sources and contract/pin documents;
- bound input artifacts with SHA-256 and resolved repository locations;
- predecessor packet references and literal authenticated packet calls;
- static gate expressions with source file and line number;
- saved boolean predicates and recorded status/scope;
- output-record locations and whether the packet is in the selected binding-reference closure.

The catalog contains 9365 static gate excerpts. This extraction parses Python syntax without importing producers. It recognizes selected gate helpers and `assert`; it does not prove completeness of gate coverage, execute dynamically generated labels, interpret every helper function, or establish that a source envelope bounds every original action term. Read the listed source and mathematical derivation for those questions.

The selected binding closure has 137 top-level packets. It includes inherited controls, alternative cases and provenance; it is intentionally broader than a minimal mathematical dependency graph. BINDING files often flatten their entire ancestor closure, so an edge in that catalog does not necessarily mean a direct numerical input. The functional figure and stage cards identify the interpreted scientific

consumption; exact binding lists identify what historical executions authenticated. Both views are needed.

Path relocation is an inspection crosswalk. The catalog first maps a historical `experiments/` suffix into the public archive and compares its manifest digest; otherwise it looks for a byte-identical archived copy. A digest-matched copy resolves content identity, not a unique historical execution. System executables, libraries, outside-line files and unresolved locators remain explicitly unembedded. No missing dependency is silently assigned a plausible file. Frozen records are never rewritten to pretend that an old execution occurred at its new location.

The archive contains failed initial implementations, resource stops, technical repairs, probes, alternate bounds, earlier radii and superseded accepted readouts. These remain visible with their own status and outputs. A failed containment is not evidence of no solution. A technical completion is not scientific acceptance. An internal review's original scope is not expanded by carrying its file into this release. The latest accepted pointer selects the operative ninth-order result without deleting predecessors.

8 Reproduction levels and commands

Run commands from the public repository root. Use a writable directory **outside** the frozen archive for reports. The following commands were chosen for record inspection; they do not execute scientific producers.

8.1 Level 0 — obtain the actual data

```
git lfs install
git lfs pull
git rev-parse HEAD
git status --short
```

Record the checked-out commit and any changes, together with the published supplement/data release identifier. For the data snapshot discussed here, use the full base commit above. A later publication commit should retain that archive and identify its own generated additions. LFS pointer retrieval may require substantial disk/network transfer; the L001 experiment file payload alone is about 14.04 GB in decimal units. This is file payload, not a measured fresh-clone disk or transfer requirement; deduplicated Git LFS storage differs.

8.2 Level 1 — file and saved-record inspection

```
python3 src/verify_l001_archive.py \
  --data-dir data/raw/l001 \
  --manifest data/processed/l001/ARCHIVE_MANIFEST.json \
  --output /tmp/l001-archive-check.json

python3 src/verify_saved_records.py \
  --data-dir data/processed/neutral-certificates \
  > /tmp/neutral-record-check.json
```

The archive checker requires only the Python standard library and streams file hashes. It rejects missing, changed, unexpected and invalid manifest paths; absent empty historical directories are reported separately. It does not use the catalog's inferred closure as a substitute for the full manifest.

The neutral checker requires Python 3.9 or later and standard-library rational/integer, gzip and hashing modules. Its expected report is PASS, **822 checks**, **16,365 saved kinetic cells**, zero new integrations and zero producer imports. It checks source-copy identities, exact saved Newton inclusion/linkage,

strict phase bands, contiguous uncertain-event covers, opposite mass endpoint signs, the kinetic rational bound and stored hull reconstruction. It does not re-prove the producers' omitted interval predicates.

The anonymous inspection package's own `README.md` describes an original standalone invocation. In this repository, use the top-level `src/verify_saved_records.py` command above, which accepts an explicit data directory. Their interfaces differ even though their saved-check purpose is closely related.

8.3 Level 2 — bounded algebra and catalog reconstruction

```
python3 tests/check_local_algebra.py \
  --output /tmp/local-algebra-check.json

python3 src/build_pipeline_catalog.py \
  --output-dir /tmp/pipeline-catalog
```

The algebra checker requires SymPy. It checks the charged trace divisor, source compatibility and neutral-time canonical density/form scaling in its displayed expressions. Its seven boolean checks do not prove the analytic source estimates or all nonlinear equations. Record Python and SymPy versions; no environment lock for the entire historical computation is claimed.

The catalog generator requires modern Python with `sys.set_int_max_str_digits` support (Python 3.11 or later). It reads the frozen manifest, terminal/results and binding files, parses top-level Python sources, and writes derived JSON/Markdown outside the raw tree. It imports no archived scientific module. Output paths must be writable; large exact-rational JSON records can require appreciable memory. Rebuilding the catalog is documentary regeneration, not regeneration of any numerical certificate.

`src/check_pipeline_documentation.py` checks the interpreted stage map, exact selected constants, saved final gate ratios, document locators and catalog linkage. Its report states its scope explicitly. It supplements the archive checker; it does not replace the seven-source or analytic-inverse audit.

8.4 Level 3 — fresh scientific regeneration

No single fresh-checkout command currently regenerates every scientific producer in this paper. The historical scripts are supplied as inspectable source, with their original contracts and records. Several use absolute Linux paths, bound native libraries, fixed historical execution windows, one-shot immutable output directories and shared lease modules. Running them directly in a completed packet can fail or attempt to create new failure records. They are not the commands for Levels 1–2.

A qualified regeneration should create a new run directory and an explicit relocation adapter, retain pristine sources, compare exact inputs/parameters, bind the new runtime, and execute the predeclared controls before production. Precision, steps, event coverage, gate inequalities and stopping conditions must remain explicit. Any mathematical or algorithmic change belongs to a separate run with a new contract and comparison, not a rewritten historical identity. Never disable a failed gate merely to finish an example.

Regeneration records should include the command, working directory, exact input/source hashes, Python and numerical library/build identity, precision/thread settings, initial state, integration domain, checkpoint protocol, terminal classification, adverse results and independent postrun disposition. A fresh output is first producer-provisional. Existing internal acceptance documents are evidence about their historical runs, not automatic approval of a new environment.

9 Assurance limits and completion checklist

The historical Arb routes share Arb, some inputs and mathematical reductions. Separate implementations help detect certain implementation errors but do not provide arithmetic independence. The kinetic

integer route uses different arithmetic for a different premise; it does not independently recertify the Arb phase or source trajectories. Runtime witnesses identify loaded files at their stated boundaries, rather than continuously proving correct execution.

The present archive permits substantially more scrutiny than a paper containing only plots or terminal decimal values. It still does not export every Picard box, Taylor/remainder array or elementary-function proof object for every foundational run. Accordingly, a reader can verify saved records and inspect the actual algorithms, but cannot reconstruct every hidden interval step from the compact outputs alone. The article states the remaining faithful-execution/outward-arithmetic premises.

The following items remain before claiming a fully portable public reproduction:

Item	Current status and required completion
Publication commit and permanent identifier	Scientific archive is pinned; this supplement and revision 6 publication assets are local additions until committed/published. Record an immutable release identifier when available.
Anonymous referee route including nonlinear evidence	The older anonymous inspection package covers B1–B4. It does not yet include the complete N1–N9 chain. A public owner-identifying archive and an anonymous package serve different access needs.
Fresh-checkout producer execution	Not qualified end to end. Supply explicit adapters, dependencies and new run contracts; execute and independently assess the resulting chain.
Runtime and external source closure	Historical binding records include system/runtime and outside-line references. Catalog resolution is partial; inventory unresolved entries, supply needed lawful dependencies or document their exact acquisition/build route.
Figure regeneration assets	Current article renderer/CSV are preserved privately. Deposit reviewed public pipeline copies and bind them to the figure outputs if public figure regeneration is promised.
Sharpened source/inverse audit	Focused scientific review must reconstruct equation-to-envelope domination and the compatible analytic inverse; terminal arithmetic is insufficient.
Archival proof objects / different arithmetic	Not supplied as a new assurance layer. Such work would strengthen scrutiny but must be distinguished from the stated current premises.

This supplement is an operational evidence guide, not an additional scientific acceptance disposition. Its documentary checks establish links, stored identities and reported arithmetic at the level stated. The article's conditional mathematical claims and the remaining scientific review priorities are unchanged.